EART27201 Sedimentary Rocks and Fossils



Macrofossils: Introduction

Why?

To function as an earth scientist you need to know about fossils

Aim

Identify, in a hand specimen or under a microscope, the major groups of micro and macro fossils, and develop the basic skills required to infer from these the ages of rocks and their palaeoenvironments.

How?



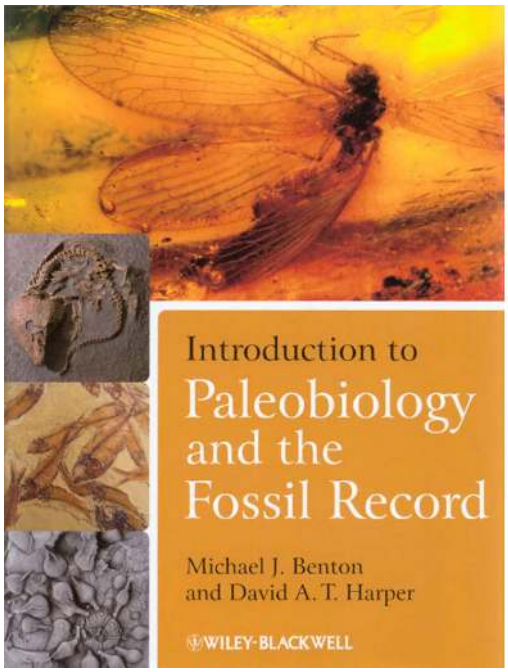
Joseph Wright of Derby – The Alchemist Discovering Phosphorus

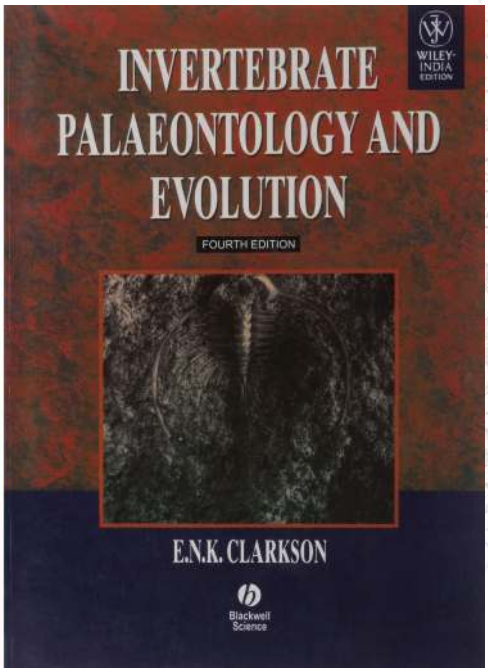
And assessment?



Joseph Wright of Derby – The Alchemist Discovering Phosphorus

Where can I learn more?





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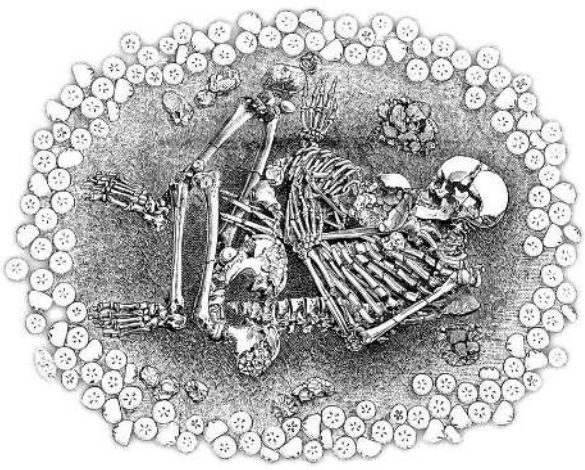
Macrofossils: History

What is a fossil?



Fossil: The remains of a once-living organism. [...] The term includes skeletons, tracks, impressions, trails, borings, and casts. Fossils are usually found in consolidated rock, but not always (e.g. woolly mammoths living 20 000 years ago were recovered from the frozen tundra of Siberia). In its original sense, fossil meant anything dug up from the earth, including ores, precious stones, etc. The modern use of the word dates from the late 17th century.
Source: Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press.
Image by H. Zell (left) & Curtis Davies (right).

Have we always known about fossils?







Taylor, P.D., 1998. Fossils in folklore. Geology Today 14, 142-145. doi:10.1046/j.1365-2451.1998.014004142.x

When Whitby's nuns exalting told, Of thousand snakes, each one Was changed into a coil of stone, When Holy Hilda pray'd: Themselves, without their holy ground, Their stony folds had often found. – Marmion, Sir Walter Scott

When was the science born?



Palaeontology: Study of the fossil remains of plants and animals. Evidence from fossils is used in the reconstruction of ancient environments and in tracing the evolution of life.

Source: Oxford University Press World Encyclopedia. DOI: 10.1093/acref/9780199546091.001.0001



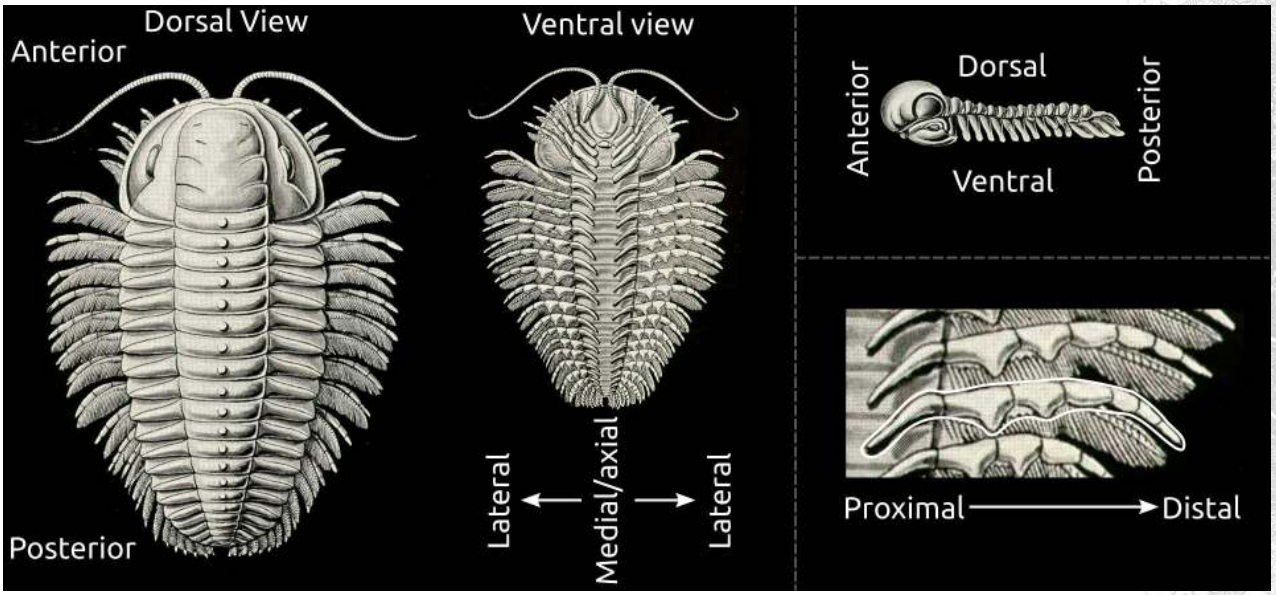
Sesquipedalian /ˌsɛskwɪpɪˈdeɪlɪən/ : adjective FORMAL (of a word) polysyllabic; long. Characterized by long words; long-winded.
e.g. "the sesquipedalian prose of scientific journals"

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Macrofossils: Some Biology

How do we describe directions on an animal?



Anatomical terms of location

Sources: Hine, R., 2015. A dictionary of biology. Oxford University Press; Kent, M., 2006. The Oxford Dictionary of Sports Science & Medicine. Oxford University Press.

Anterior: Designating the part of an animal that faces to the front, i.e. that leads when the animal is moving.

Posterior: Designating the part of an animal that is to the rear, i.e. that follows when the animal is moving.

Dorsal: Describing the surface of a plant or animal that is farthest from the ground or other support, i.e. the upper surface. In vertebrates, the dorsal surface is that down which the backbone runs.

Ventral: Describing the surface of a plant or animal that is nearest or next to the ground or other support, i.e. the lower surface. In bipedal animals, such as humans, it is the forward-directed (anterior) surface.

Median/axial: Situated in the middle or oriented towards the midline; the opposite of lateral.

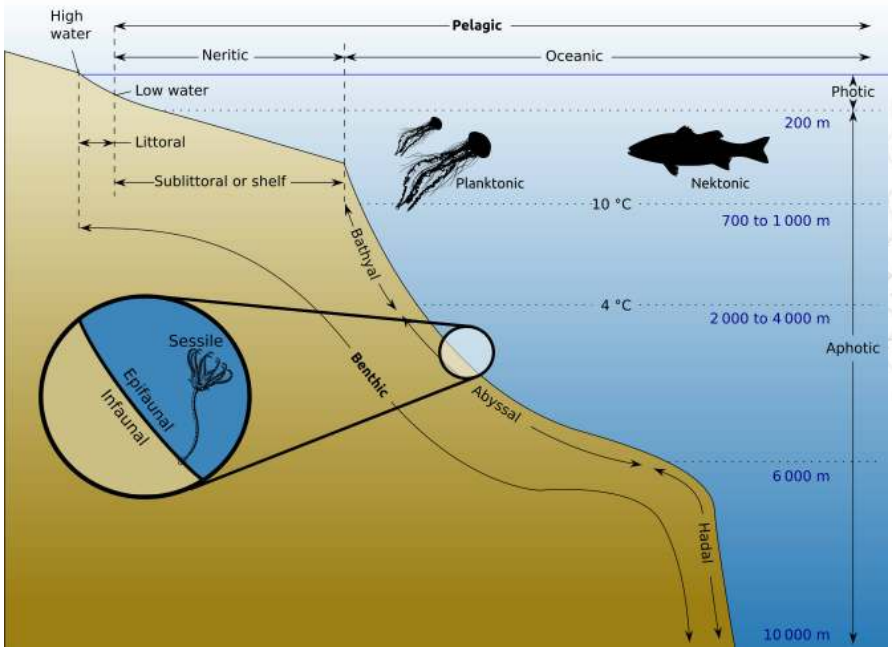
Lateral: In anatomy, applied to structures away from the midline or on the outer surface of the body.

Proximal: Closer to the trunk (e.g. the elbow is proximal to the wrist) or the origin of a structure.

Distal: Describing the part of an organ that is farthest from the organ's point of attachment to the rest of the body. For example, hands and feet are at the distal ends of arms and legs, respectively. Compare proximal.



How do we describe where and how animals live?



Pelagic: Describing organisms that swim or drift in a sea or a lake, as distinct from those that live on the bottom (see benthos). Pelagic organisms are divided into plankton and nekton.

Benthic: Occurring on the bottom of a sea or lake. Benthic organisms may crawl, burrow, or remain attached to a substrate.

Plankton (Planktonic): The collection of pelagic organisms that drift or float passively with the current in a sea or lake. Plankton forms an important food source for many other members of the aquatic community and is divided into zooplankton and phytoplankton.

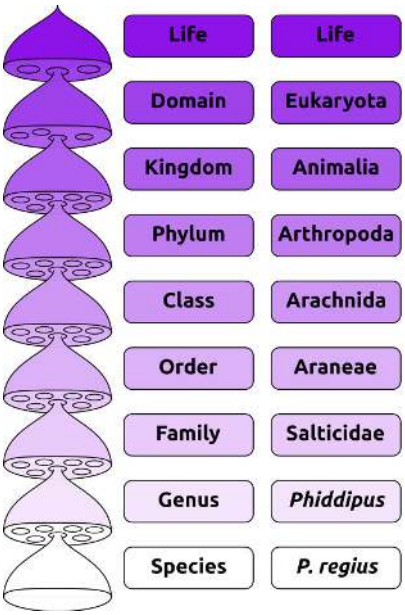
Nekton (Nektonic): Pelagic organisms that actively swim through the water.

Epifaunal: Applied to benthic organisms that live on the surface of the sea bed, either attached to objects on the bottom or free-moving. They are characteristic of the intertidal zone.

Infaunal: Applied to benthic organisms that dig into the sea bed or construct tubes or burrows. They are most common in the subtidal and deeper zones (i.e. the area seaward of the low-water mark).

Sessile: Describing animals that live permanently attached to a surface, i.e. sedentary animals. Many marine animals, e.g. sea anemones and limpets, are sessile.

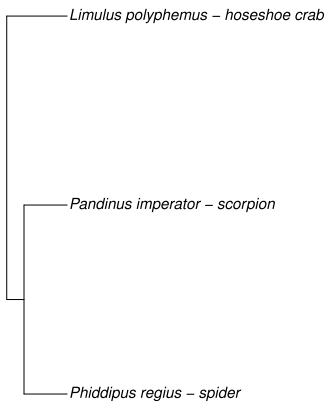
How do we organise life?

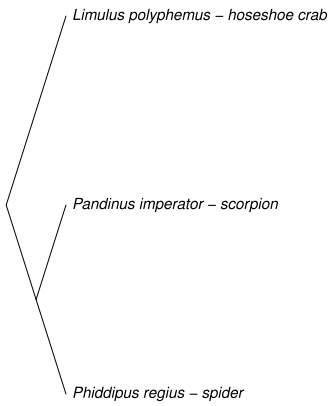




Taxonomy: The scientific classification of organisms. By extension, the term is also given a wider use (e.g. soil taxonomy). The word is derived from the Greek tasso, meaning 'arrange'.

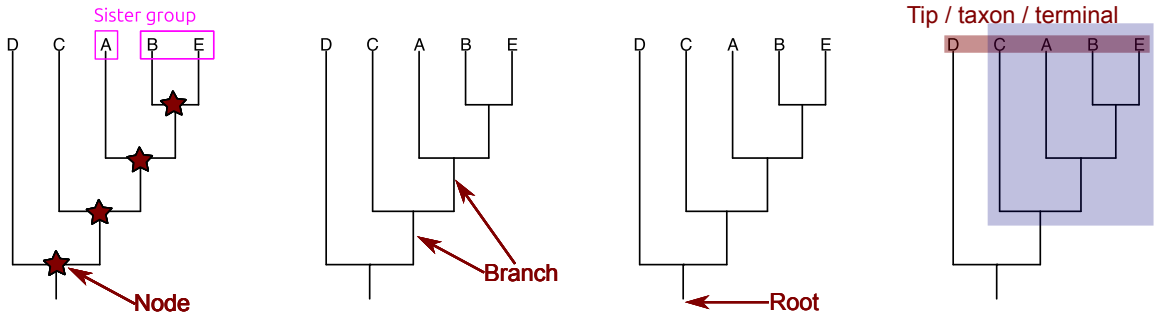
Source: Hine, R. 2019. A Dictionary of Biology (8th Edition). Oxford University Press. | Jumping spider © Thomas Shahan (CC BY-NC-ND)

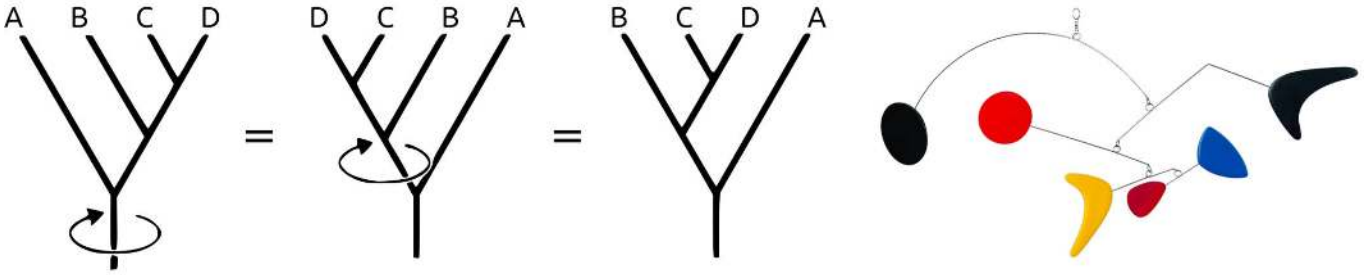




Clade: A group of organisms comprising an ancestor and all of its descendant forms.
Phylogeny: The evolutionary history of an organism or group of related organisms (a clade).

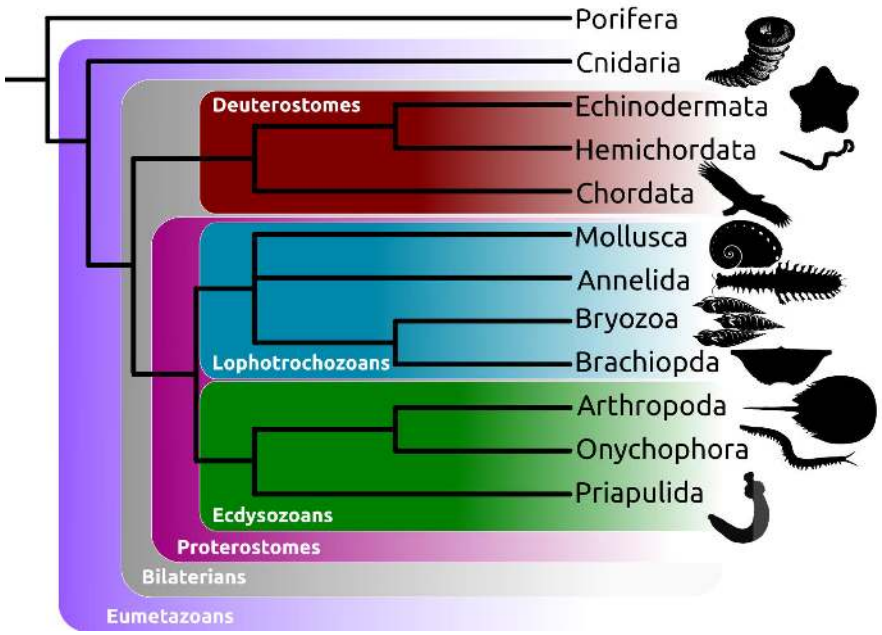
```
#R code used to create these trees:
library(ggtree); library(ggplot2); library(ape); tree <- read.tree(text =
"((Phiddipus_regius_-_spider:5, Pandinus_imperator_-_scorpion:5):2,Limulus_polyphemus_-_hoseshoe_crab:7);");
layout(matrix(1:2, 1, 2, byrow = TRUE)); plot(tree,label.offset = 0.2); plot(tree, "c", FALSE, label.offset = 0.2);
```



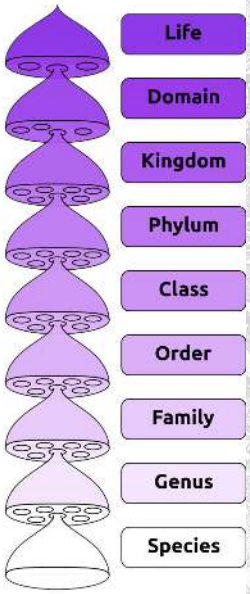


Right: FloatingArtMobiles

Phylogeny

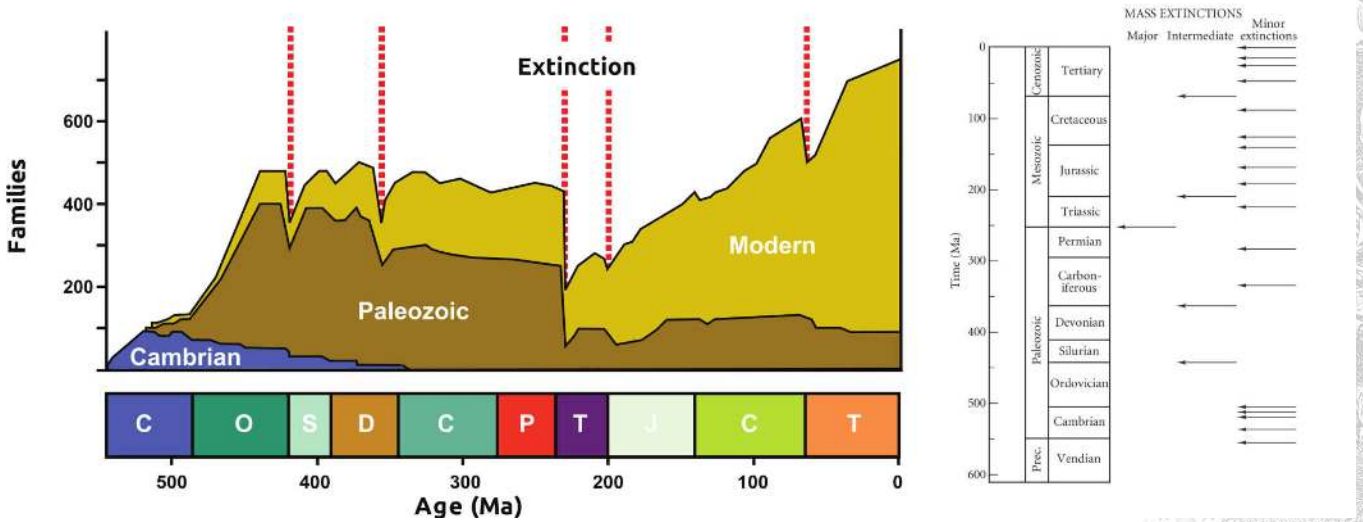


Taxonomy

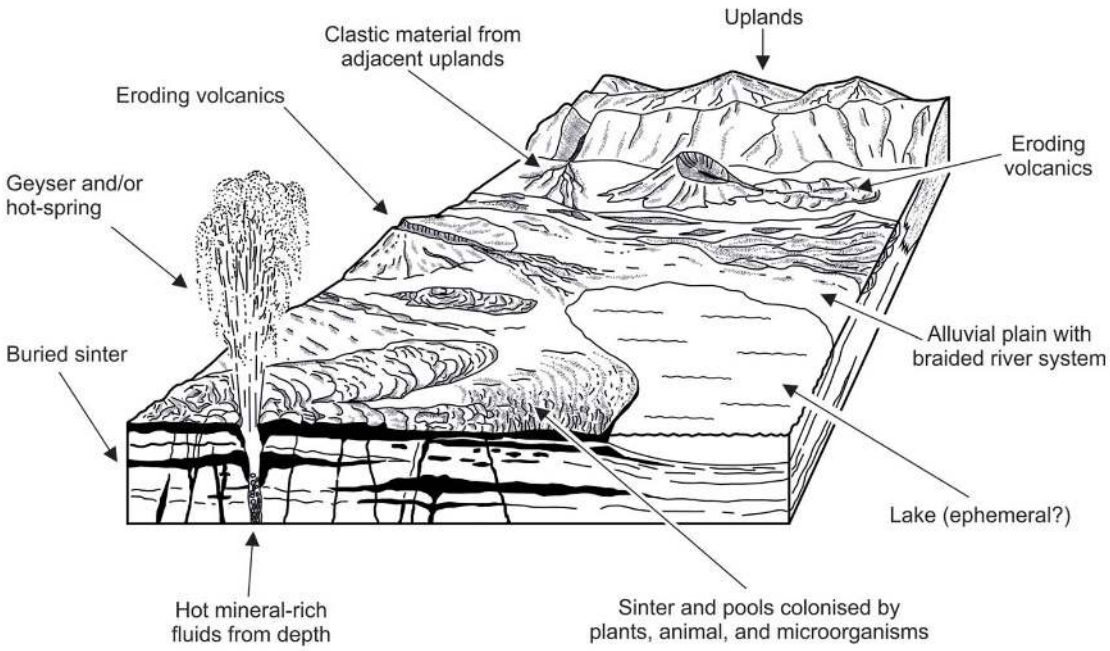


The utility of fossils

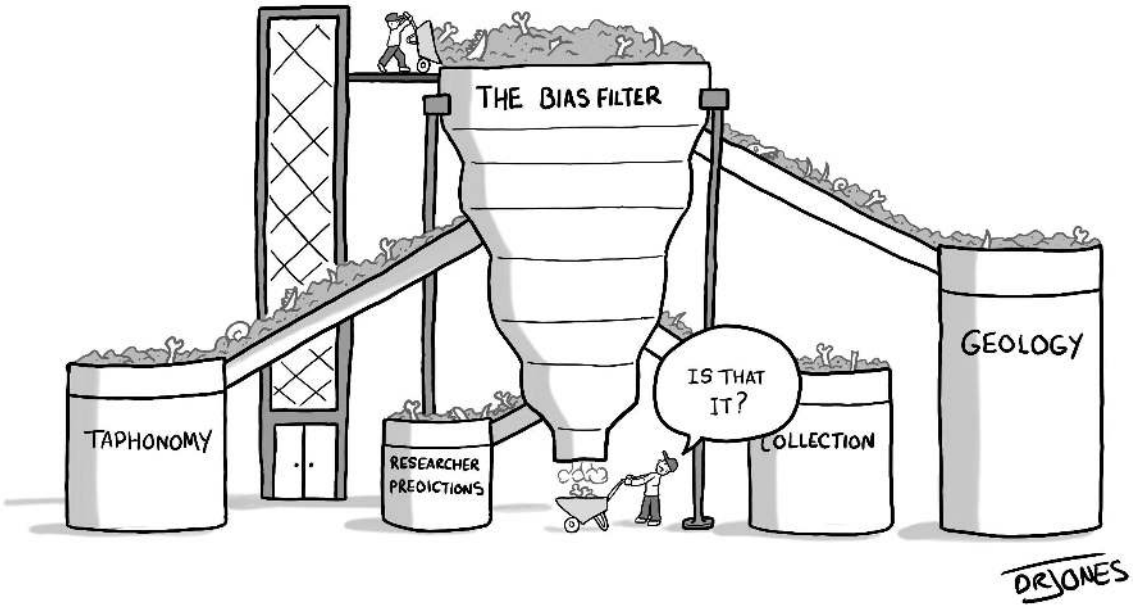
How do fossils help geologists?



Jack Sepkoski (1948-1999). An overview - Sepkoski, J.J., 1997. Biodiversity: past, present, and future. Journal of Paleontology, 71(4), pp.533-539. Left, image modified after: Servais, T., Harper, D.A., Munnecke, A., Owen, A.W. and Sheehan, P.M., 2009. Understanding the Great Ordovician Biodiversification Event (GOBE): Influences of paleogeography, paleoclimate, or paleoecology. GSA Today, 19(4), pp.4-10.
Right: Benton, M. and Harper, D.A., 2009. Introduction to paleobiology and the fossil record. John Wiley & Sons.



Source: Garwood, R.J., Oliver, H. and Spencer, A.R., 2020. An introduction to the Rhynie chert. Geological Magazine, 157(1), pp.47-64.



Cartoon by Dr Ellis Jones, Ratbot Comics

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Macrofossils: Taphonomy

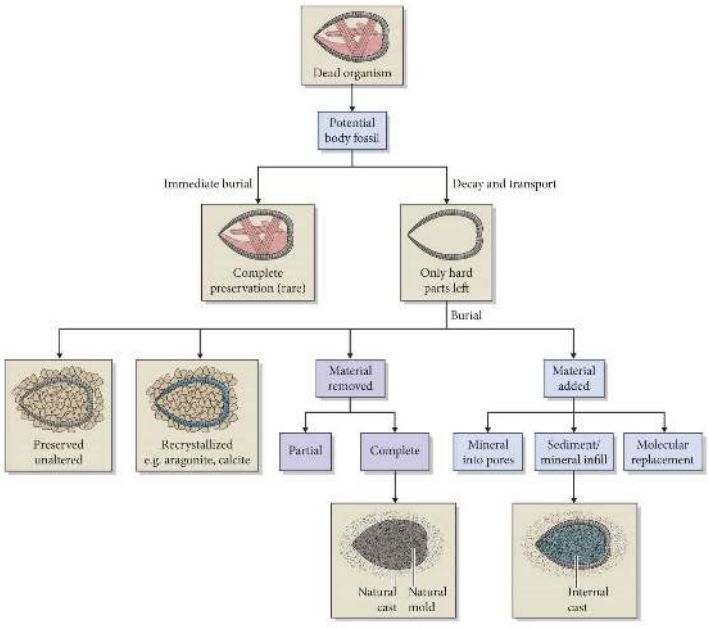
Taphonomy: The study of the transition of all or part of an organism and its traces from the biosphere into the lithosphere (i.e. fossilization).
Source: Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press.

What are animals made of?

	Inorganic				Organic				
Group	Carbonates	Calcite	Phosphates	Silica	Iron oxides	Chitin	Cellulose	Collagen	Keratin
Prokaryotes									
Algae									
Higher plants									
Protozoa									
Fungi									
Porifera									
Cnidaria									
Bryozoa									
Brachiopoda									
Mollusca									
Annelida									
Arthropoda									
Echinodermata									
Chordata									

Dark red = common; light red = lesser occurrences
After Benton, M. and Harper, D.A., 2009. Introduction to paleobiology and the fossil record. John Wiley & Sons.

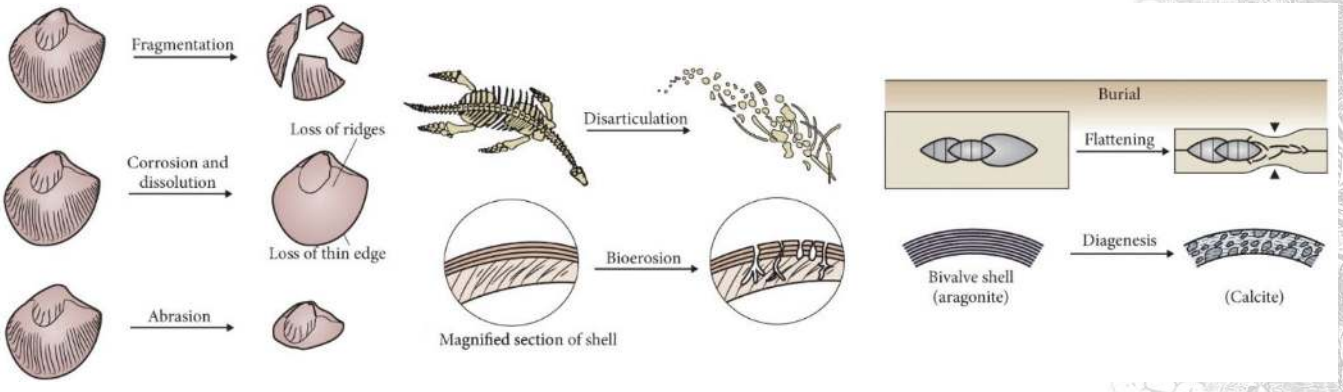
What happens after something dies?



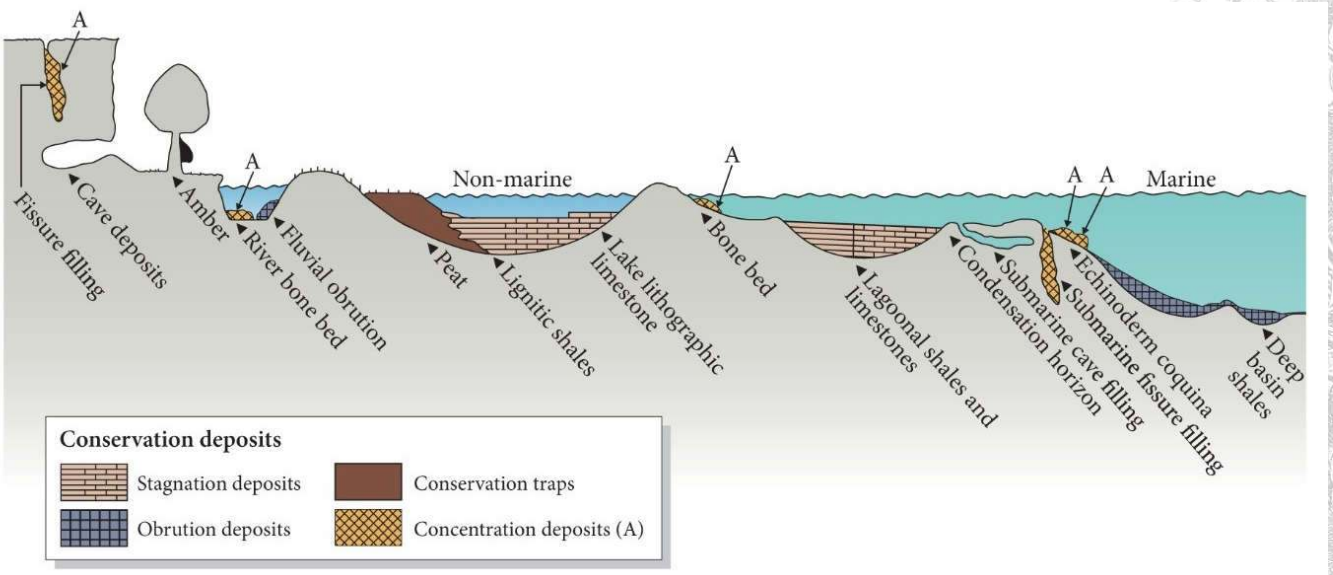
Source: Benton, M. and Harper, D.A., 2009. Introduction to paleobiology and the fossil record. John Wiley & Sons.



Image by Sarah Murray.

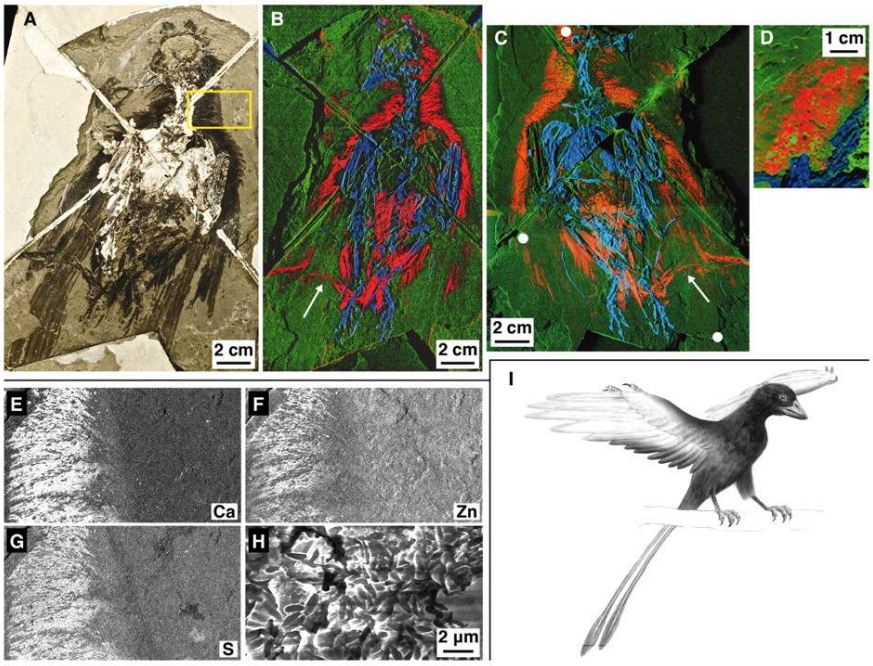


Source: Benton, M. and Harper, D.A., 2009. Introduction to paleobiology and the fossil record. John Wiley & Sons.



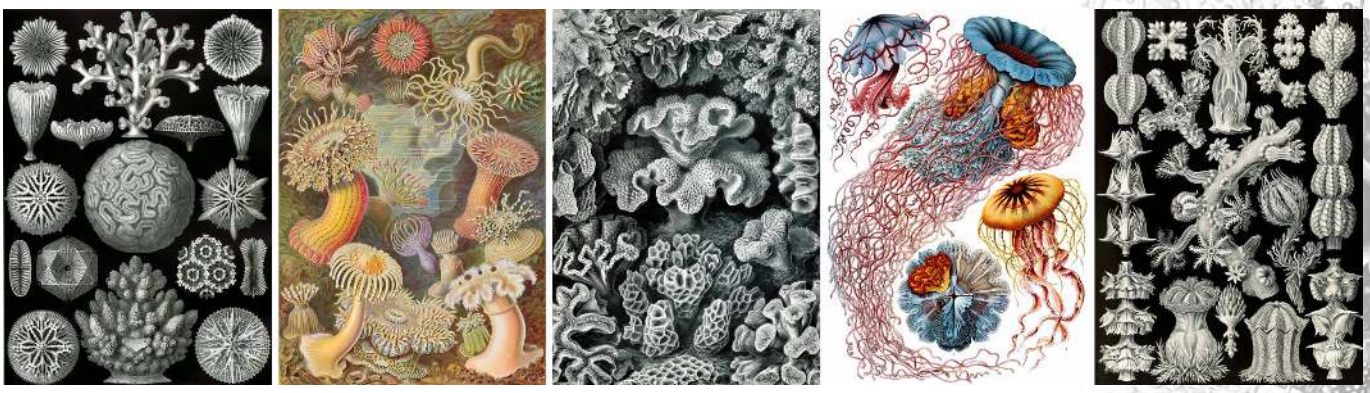
Lagerstätte (pl. Lagerstätten): A sedimentary deposit that is of value because of the fossils it contains. A fossil-Lagerstätte contains an abundance of fossils; a conservation-Lagerstätte is a stratum in which fossils are exceptionally well preserved; a concentration-Lagerstätte is a large accumulation of fossils of particular species, such as a shell bed.

Source: Source: Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press.



Confuciusornis sanctus. Source: Wogelius, R.A., Manning, P.L., Barden, H.E., Edwards, N.P., Webb, S.M., Sellers, W.I., Taylor, K.G., Larson, P.L., Dodson, P., You, H. and Da-Qing, L., 2011. Trace metals as biomarkers for eumelanin pigment in the fossil record. Science, 333(6049), pp.1622-1626.

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Macrofossils: Corals – Introduction and biology

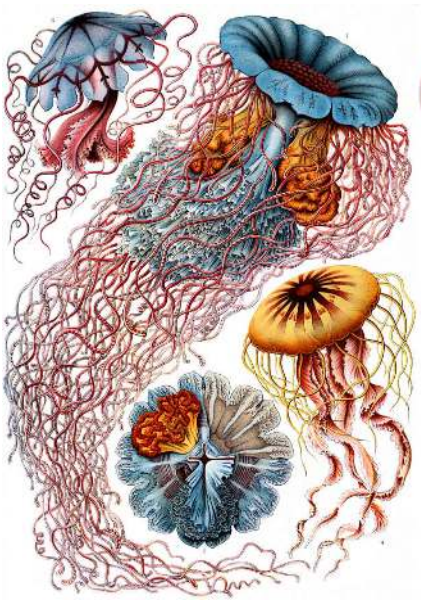
Russell Garwood | russell.garwood@manchester.ac.uk

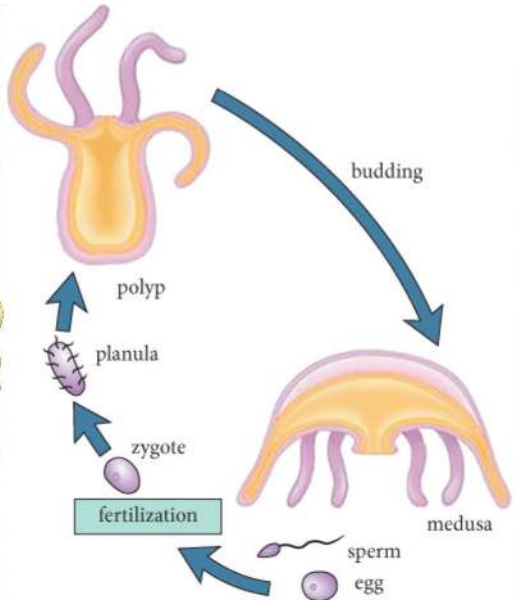
What is a Coral?



Image, middle: The Digital Atlas of Ancient Life. Left, Right: Hackel, E. 1899. Kunstformen der Natur.

What is a Cnidarian?

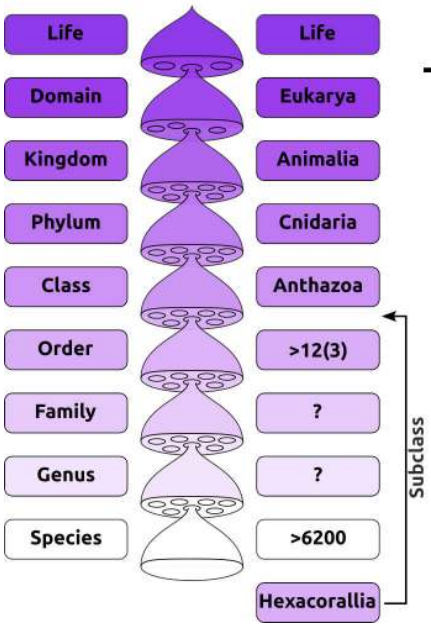


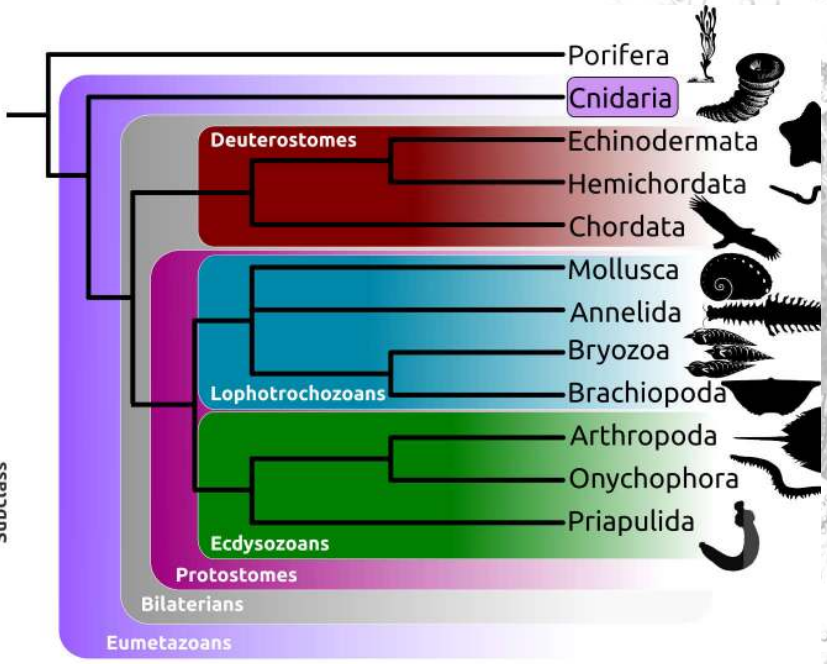




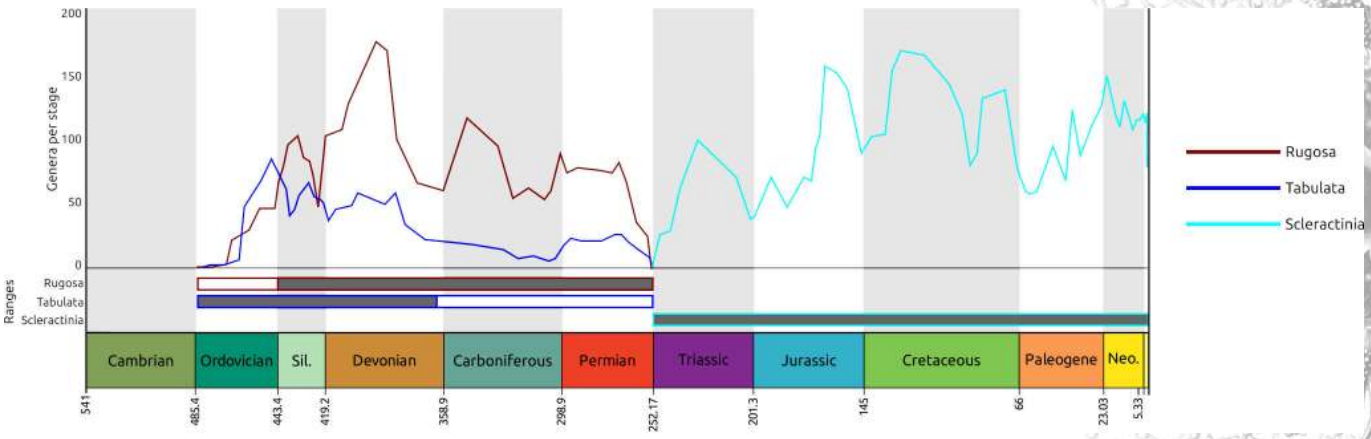
Souce: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

Where do they sit on the tree of life?





How are corals distributed through time?



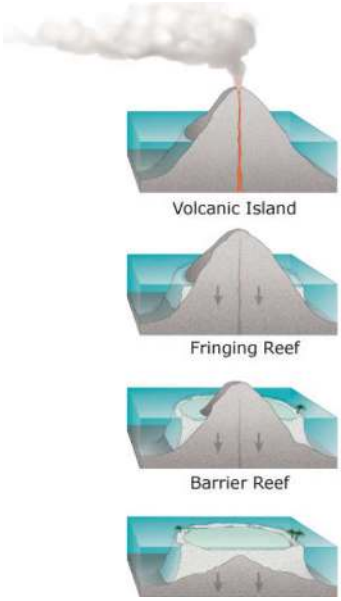
Fossil data from PBDB.

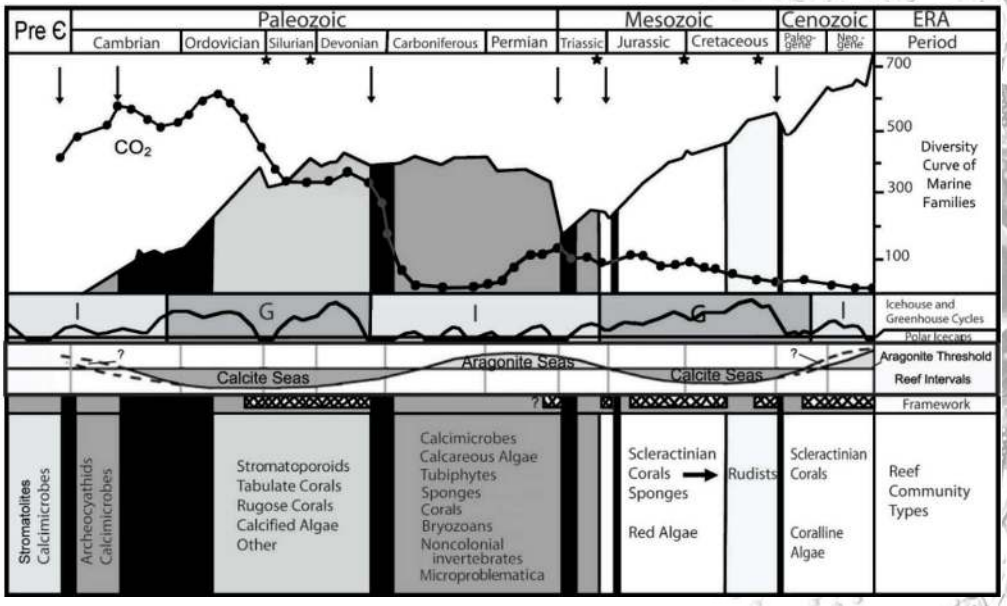
Why do they matter?



Left: Nick Hobgood; Middle: Toby Hudson; Right: Jim Maragos.

How are reefs distributed through time?





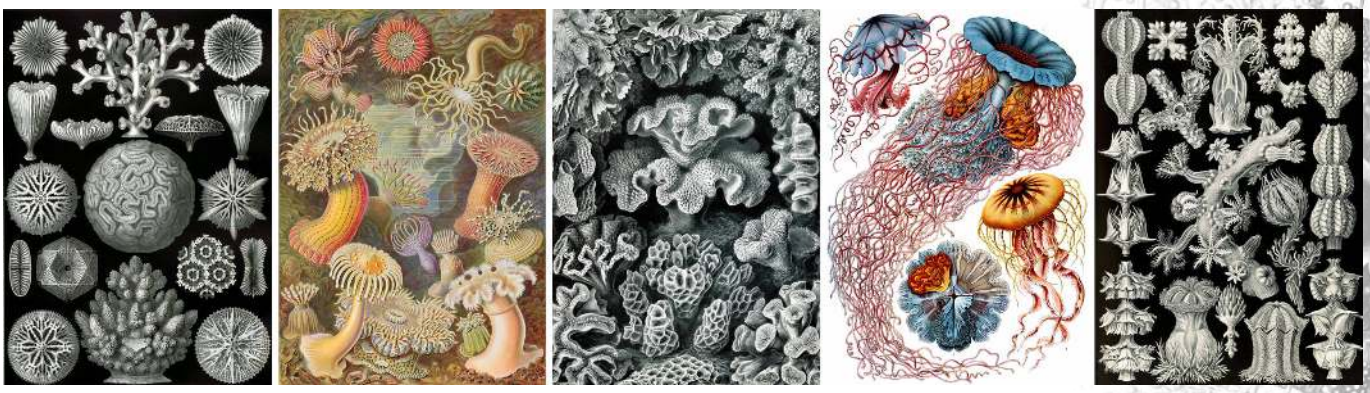
Left: USGS. Right: Stanley, G.D. and Lipps, J.H., 2011. Photosymbiosis: the driving force for reef success and failure. The Paleontological Society Papers, 17, pp.33-59.

Why do they matter 2?



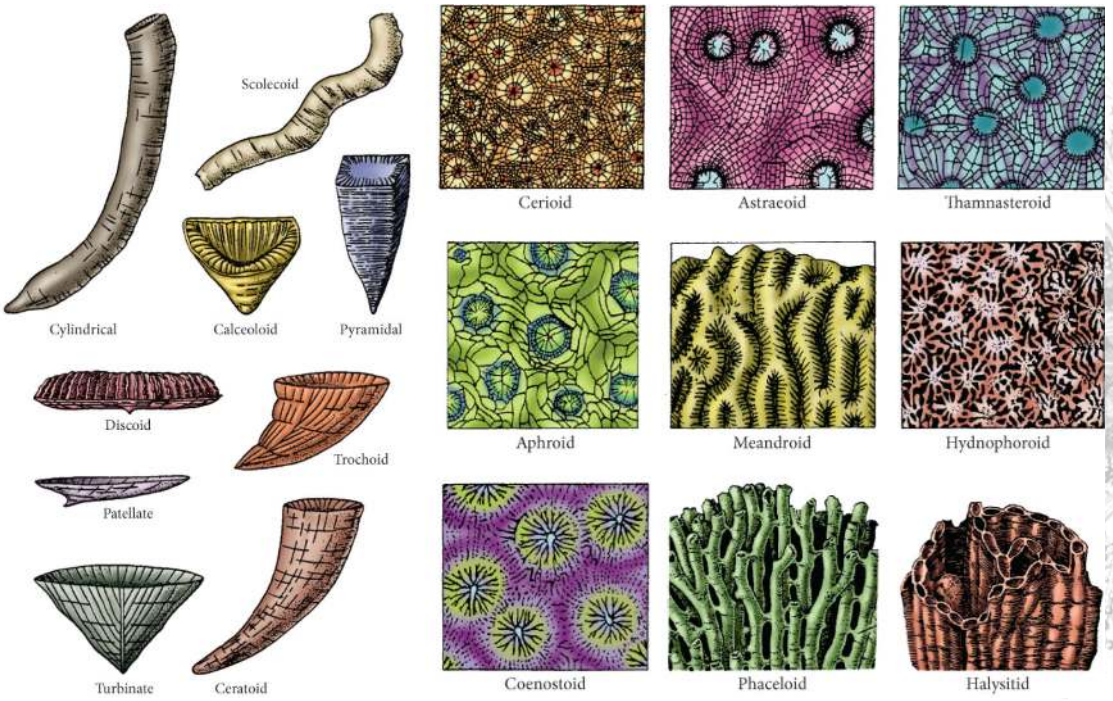
Photograph by James L. Stuby.

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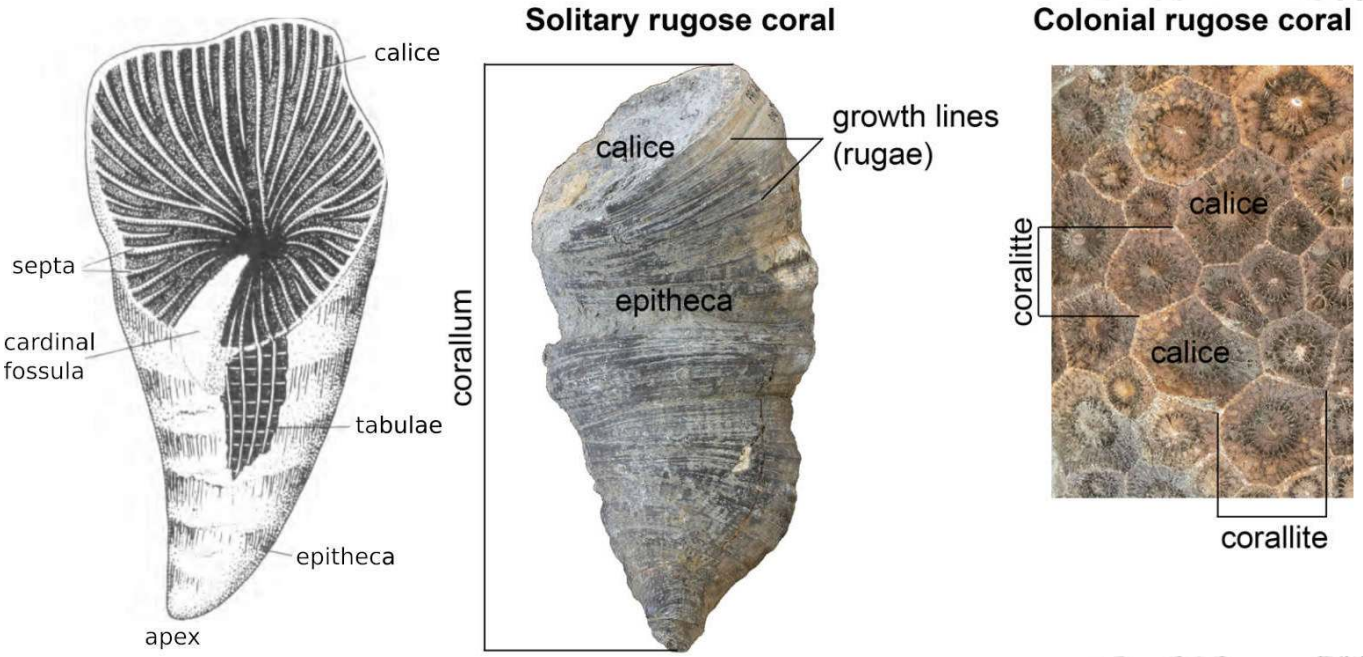


Macrofossils: Corals – Morphology / Anatomy

Russell Garwood | russell.garwood@manchester.ac.uk



Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.



Left: Clarkson, E.N.K., 2009. Invertebrate palaeontology and evolution. John Wiley & Sons. Right: Digital atlas of ancient life.

Corralum: A solitary coral individual, or the whole skeleton of a colonial coral, made up from individual skeletons (corallites).

Septum (pl. septa) "fence": A dividing wall within the skeleton of various animals (adj. septate). Here vertical (i.e. up-down) partitions arranged like spokes of a wheel.

Theca "case": the skeletal wall of a coral. Theca may in turn be covered by an outermost skeletal sheath called the epitheca.

Ruga (pl. rugae) "roughness": Irregular small projections (adj. rugose). In corals these are on epithica and are ~growth lines.

Apex: "tip" or top (adj. apical).

Calice: The upper part of the skeleton of a coral (the corallum), in which the polyp sits.

Tabula (pl. tabulae) "table": a horizontal division within the skeleton - a sheet secreted by polyp as coral grows.

Dissepiment "partition": a horizontal, or nearly horizontal, plate of tissue supporting tabulae.

Axial structure: Septa fuse in the middle - the axial structure - which occupies core region.

How do the major groups differ?

Feature	Rugosa 	Tabulata 	Scleractinia 
Growth mode	Colonial and solitary	Colonial	Colonial and solitary
Septa	6 prosepta; later septa in only 4 spaces	Septa weak or absent	6 prosepta; later septa in all 6 spaces
Tabulae	Usual	Well developed	Absent
Skeletal material	Calcite	Calcite	Aragonite
Stability	Poor	Poor	Good with basal plate
Range	Ordovician to Permian	Ordovician to Permian	Triassic to Recent

Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.